

The empirical recurrence for OEIS A154145, proved from an exact model

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Abstract

OEIS A154145 carries an empirical recurrence of order 9. It is true, and it is decidable rather than empirical. The entry's sequence is given exactly by the solutions of a Pell equation, in their finitely many orbits, from which a provably satisfies a MONIC linear recurrence of order $S = 10$, derived below and not assumed. The residual of the conjectured recurrence satisfies the same one, so whether it annihilates a is settled by a finite exact computation. No transfer matrix is involved: the sequence is the ordered list of solutions of a Pell equation, and what makes it C -finite is the automorph of the quadratic form.

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1 The conjecture

OEIS A154145 is “Indices k such that 15 plus the k -th triangular number is a perfect square.”. It has offset 1 and begins

$$1, 4, 6, 11, 20, 33, 43, 70, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

Conjectures: (Start) $a(n) = +a(n-1) + 6a(n-4) - 6a(n-5) - a(n-8) + a(n-9)$.

As of the “Last modified” line on the live entry (Dec 23 2024 14:53:42, revision 14) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a Pell equation

$k(k+1)/2 + 15$ is a perfect square exactly when 8 times it is, and completing the square turns that into

$$(2k+1)^2 - 2(2m)^2 = 1 - 8 \cdot 15 = -119,$$

so writing $X = 2k+1$ and $Y = 2m$ the admissible k are exactly the solutions of $X^2 - 2Y^2 = -119$ with X odd and positive and Y even and nonnegative. The entry's sequence is that solution set, listed by increasing k .

The form $x^2 - 2y^2$ has the automorph coming from the unit $3 + 2\sqrt{2}$,

$$T(X, Y) = (3X + 4Y, 2X + 3Y),$$

which carries solutions to solutions; and every solution is T^j of a FUNDAMENTAL one, meaning one whose preimage $T^{-1}(X, Y) = (3X - 4Y, -2X + 3Y)$ leaves the region. A finite search finds them – there are 4 here – and the search is checked rather than bounded by a quoted theorem: it is run to a limit and then re-run to four times that limit, and the engine refuses unless the two agree.

3 The bound

On the region $X \geq 1$, $Y \geq 0$ one has $Y = \sqrt{(X^2 - 119)/2}$, an increasing function of X , so $T(X) = 3X + 4Y$ is strictly increasing: T PRESERVES THE ORDER of solutions. Hence if the sorted list of solutions has its $(n + 4)$ -th entry equal to T of its n -th for 4 consecutive n , it does so for every later n – the 4 orbits interleave in a fixed cyclic order from there on. That happens from index $n_0 = 0$, checked directly.

Within one orbit $X_{j+2} = 6X_{j+1} - X_j$, since $3 + 2\sqrt{2}$ and $3 - 2\sqrt{2}$ are the roots of $t^2 - 6t + 1$; in terms of $k = (X - 1)/2$ that reads $k_{j+2} = 6k_{j+1} - k_j + 2$. Combining with the interleaving,

$$a(n + 2 \cdot 4) = 6a(n + 4) - a(n) + 2 \quad (n \geq n_0),$$

so $(z - 1)(z^{2 \cdot 4} - 6z^4 + 1)$ annihilates a , monic of order $S = 10$ once the transient is allowed for. Nothing in that bound is estimated: 4 is the number of fundamental solutions and n_0 is where the interleaving is observed to settle, which the order-preservation of T then makes permanent.

4 The proof

Lemma 1. *Let $A(z) = z^S - \sum_{j < S} \alpha_j z^j$ be the monic annihilator derived above and $q(z) = z^9 - \sum_i c_i z^{9-i}$ the characteristic polynomial of the conjectured recurrence. The residual $r(n) = a(n) - \sum_i c_i a(n - i)$ is a linear combination of shifts of a , so A annihilates r as well. Since $A(0) \neq 0$ the recurrence it gives runs in both directions, and S consecutive zeros of r force r to vanish at every later index.*

Proof. A annihilates a , hence every shift of a , hence any linear combination of shifts; that is r . A monic recurrence with nonzero constant term determines each term from the S before it and each from the S after it, so a run of S zeros propagates. $\square$

Theorem 2.

$$a(n) = a(n - 1) + 6a(n - 4) - 6a(n - 5) - a(n - 8) + a(n - 9)$$

for every $n > 1$.

Proof. The terms $a(0), \dots$ were computed exactly in integer arithmetic from the model, extended by A where the model itself was not evaluated, and $r(n)$ formed for each index. Those integers vanish from $n = 1 + 1$ onwards, and the run exceeds $S + 9$, which by Lemma 1 settles every larger index at once. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 37 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: it is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no model involved, and holds wherever the proved range and the data overlap.

Third, the derived annihilator A was itself tested on terms it had not been given: the model was evaluated beyond the S values that determine it and A reproduced them. A bound that is too small fails this test, which is why it is run.

References

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